

$$* f(x, y) = y \sin x$$

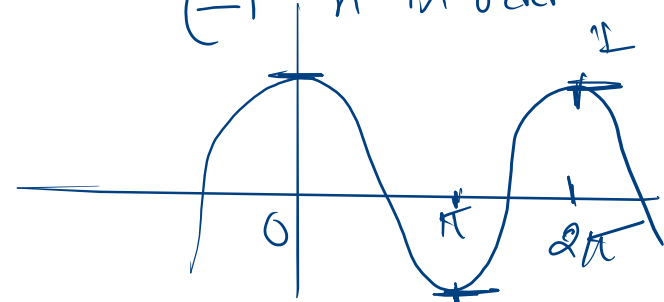
$$\cos x = 0, \Rightarrow \nabla \cos x = \cos(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

$$\cos n\pi = (-1)^n, \begin{cases} 1 & n \text{ in even} \\ -1 & n \text{ in odd} \end{cases}$$

$$f_x = y \cos x, \quad f_y = \sin x, \quad f_{xx} = -y \sin x, \quad f_{yy} = 0$$

Critical points:

$$f_{xy} = \cos x$$



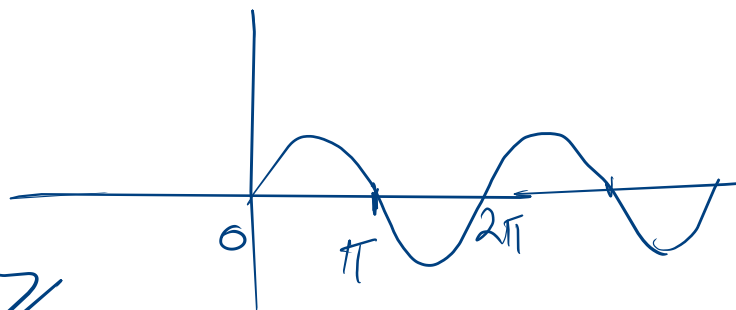
$$y \cos x = 0$$

$$\sin x = 0$$

$$\textcircled{y=0} \quad \boxed{\cos x \neq 0} \quad \times$$

$$\Rightarrow \sin x = \sin n\pi$$

$$\Rightarrow \boxed{x = n\pi}, n \in \mathbb{Z}$$



$$(n\pi, 0)$$

$$\downarrow$$

$$(x_0, y_0)$$

$$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

$$= -y \sin x - \cos^2 x$$

$$= 0 - (\cos n\pi)^2 = 0 - \{(-1)^n\}^2 = -1 < 0$$

$\delta < 0$, $(n\pi, 0)$ are all saddle points, $n \in \mathbb{Z}$.

$$f(x) \approx \underline{f(x_0)} + \underline{f'(x_0)}(x-x_0) + \underline{f''(x_0)}(x-x_0)^2 + \dots \quad \swarrow \text{Taylor}$$

$$f(x) \approx f(x_0) + \underline{f'(x_0)} \underline{(x-x_0)} + f'' \frac{2!}{2!} (x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)(x-x_0)^n}{n!}$$

Taylor Polynomial

Single Variable

one

For two variables: $f(x, y)$. $\rightarrow$ 2 variables

First degree Taylor Polynomial $\downarrow$ whose first partial derivative exists at point (x_0, y_0) , then the 1st degree Taylor polynomial of f for (x, y) near the point (x_0, y_0) is

$$f(x, y) \approx \underbrace{L(x, y)} \approx \underbrace{f(x_0, y_0)} + \underbrace{f_x(x_0, y_0)} \underbrace{(x - x_0)} + \underbrace{f_y(x_0, y_0)} \underbrace{(y - y_0)}$$

$L(x, y)$ is also called Linear approximation / Linear polynomial of f for (x, y) near (x_0, y_0) .

Taylor Polynomial (T. P.).

Second degree Taylor Polynomial (Quadratic Taylor polynomial)

For two variable function $f(x, y)$. $\frac{f_{xy}}{2}$ $\frac{f_{yx}}{2}$ $(f_{xy} = f_{yx})$
first partial derivative & second partial derivative
exist -

$$\begin{aligned} f(x, y) \approx Q(x, y) \approx & f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) \\ & + \frac{f_{xx}(x_0, y_0)}{2!}(x - x_0)^2 + \frac{f_{yy}(x_0, y_0)}{2!}(y - y_0)^2 \\ & + f_{xy}(x_0, y_0)(x - x_0)(y - y_0). \end{aligned}$$

$$\approx \underline{L(x,y)} + f_{xx}(x_0, y_0) \frac{(x-x_0)^2}{2!} + f_{yy}(x_0, y_0) \frac{(y-y_0)^2}{2!} \\ + f_{xy}(x_0, y_0) (x-x_0)(y-y_0).$$

Ex: 1 Determine the 1st & 2nd degree Taylor polynomial $L(x,y)$ & $Q(x,y)$ for the following function $f(x,y) = \sin 2x + e^{xy}$ near the point $(0,0)$. $f(0,0) = 1$.

$$f_x = 2\cos 2x, \quad f_x(0,0) = 2, \quad f_y = e^{xy}, \quad f_y(0,0) = 1$$

$$f_{xx} = -4\sin 2x, \quad f_{xx}(0,0) = 0, \quad f_{yy} = e^{xy}, \quad f_{yy}(0,0) = 1.$$

$$f_{xy} = 0$$

$$f(x, y) \approx L(x, y)$$

$$\approx f(0, 0) + f_x(0, 0)(x-0) + f_y(0, 0)(y-0).$$

$$\approx 1 + 2x \quad (\text{1st degree T.P.})$$

$$f(x, y) \approx Q(x, y) \approx 1 + 2x + \frac{f_{xx}(0, 0)}{2!}(x-0)^2 + \frac{f_{yy}(0, 0)}{2!}(y-0)^2 + f_{xy}(0, 0)(x-0)(y-0)$$

$$\approx 1 + 2x + 0 + \frac{(-1)}{2}y^2 + 0$$

$$\approx 1 + 2x - \frac{1}{2}y^2$$

Exo 2 Determine Linear & Quadratic Taylor Expansion
for $f(x, y) = x\sqrt{y}$, $P(\underbrace{1, 4}_{x_0, y_0})$.

$$f(1, 4) = 1\sqrt{4} = 2 \quad f_x = \sqrt{y}, \quad f_x(1, 4) = 2. \quad f_{xx} = 0$$

$$f_{xy} = \frac{1}{2\sqrt{y}}, \quad f_{xy}(1, 4) = \frac{1}{4}. \quad f_y = \frac{x}{2\sqrt{y}}, \quad f_y(1, 4) = \frac{1}{4}.$$

$$f_{yy} = \frac{x}{2} \left(-\frac{1}{2}\right) \frac{1}{y^{3/2}} = \frac{-x}{4y^{3/2}} \quad f_{yy}(1, 4) = \frac{-1}{4 \cdot (2^3)^{1/2}} = \left(-\frac{1}{32}\right).$$

$$L(x, y) \approx f(1, 4) + f_x(1, 4)(x-1) + f_y(1, 4)(y-4).$$

$$\approx 2 + 2(x-1) + \frac{1}{4}(y-4) \approx 2x + \frac{1}{4}y - 1$$

(Verify).

$$Q(x,y) \approx 2x + \frac{1}{4}y - 1 + \frac{f_{xx}(1,4)}{2!}(x-1)^2 + \frac{f_{yy}(1,4)}{2!}(y-4)^2$$

$$+ f_{xy}(1,4)(x-1)(y-4).$$

$$\approx 2x + \frac{1}{4}y - 1 + 0 + \frac{\left(-\frac{1}{32}\right)}{2!}(y-4)^2 + \frac{1}{4}(x-1)(y-4).$$

$$\approx 0$$

Ex: 1. $f(x,y) = xe^y + 1$, $(1,0)$

2. practice sheet (Assignment)

Quiz-2

1. Partial Differentiation (Chain Rule).

2. Relative Maximum, Minimum, Saddle point

3. Taylor Polynomial for 2 variable function